BLOCKCHAIN Experiment No: 4

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AIM: To develop a blockchain-powered web application using Solidity programming language on Remix IDE and embedding Metamask Wallet

THEORY:

Smart Contract:
A smart contract is a program stored on the blockchain that automatically executes predefined operations when its functions are called. In this experiment, the smart contract is written using Solidity.

Solidity:
Solidity is a programming language used to develop smart contracts for Ethereum-compatible blockchain networks.

Remix IDE:
Remix is a web-based IDE used to write, compile, deploy, and interact with Solidity smart contracts.

MetaMask:
MetaMask is a cryptocurrency wallet that allows users to manage blockchain accounts and connect decentralized applications to blockchain networks. It is used to approve and sign transactions in this experiment.

Testnet:
A testnet is a blockchain network used for development and testing. It uses test cryptocurrency, such as Sepolia ETH, which does not have real monetary value.

Transaction:
A blockchain transaction is an operation submitted to the network. Deployment and state-changing smart contract functions require a transaction and are confirmed through MetaMask.

TASKS PERFORMED

1. Installed and configured the **MetaMask Wallet.**
2. Created/imported a MetaMask account and connected it to the **Sepolia/RSK Testnet.**
3. Obtained test cryptocurrency from the testnet faucet.
4. Created a simple **Solidity smart contract** in Remix IDE.
5. Compiled and deployed the smart contract using MetaMask.
6. Interacted with the deployed smart contract and performed transactions.
7. Verified the transaction details using the blockchain explorer.

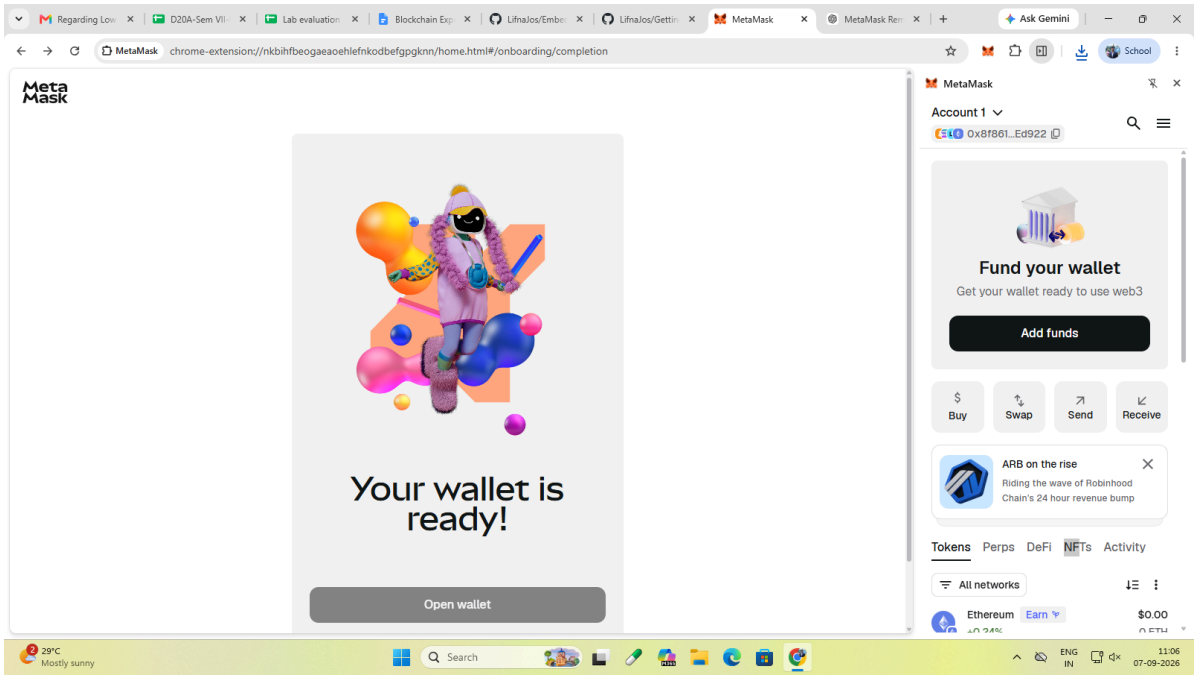
PROCEDURE:

Step - 1 : Set Up MetaMask:

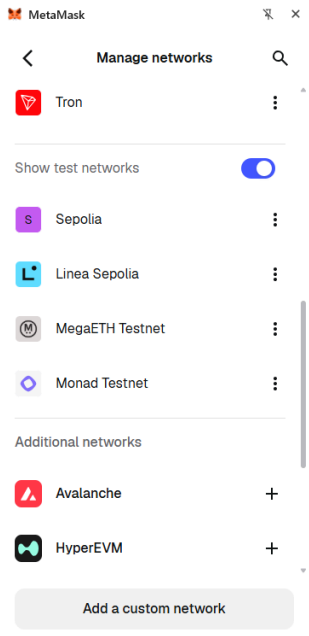
1. Install MetaMask :

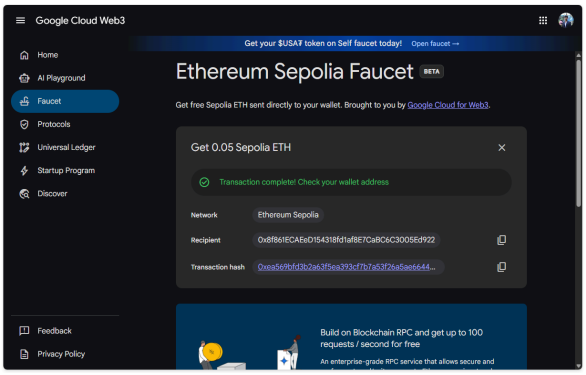
- Follow the Link to install the Google Chrome Extension.

2. Create or Import a Metamask Wallet

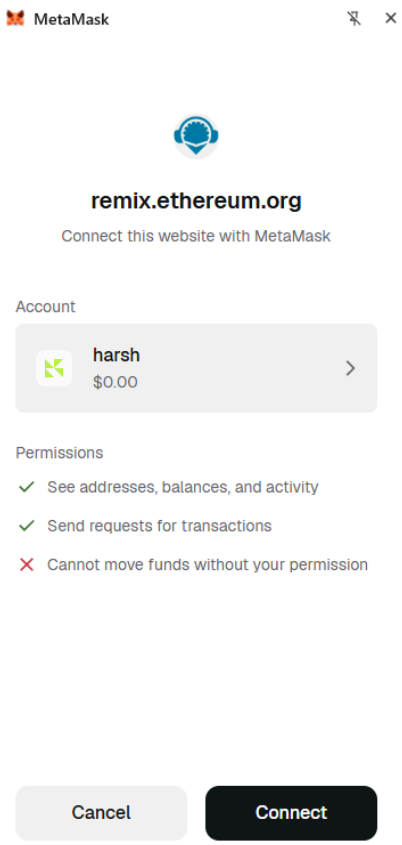


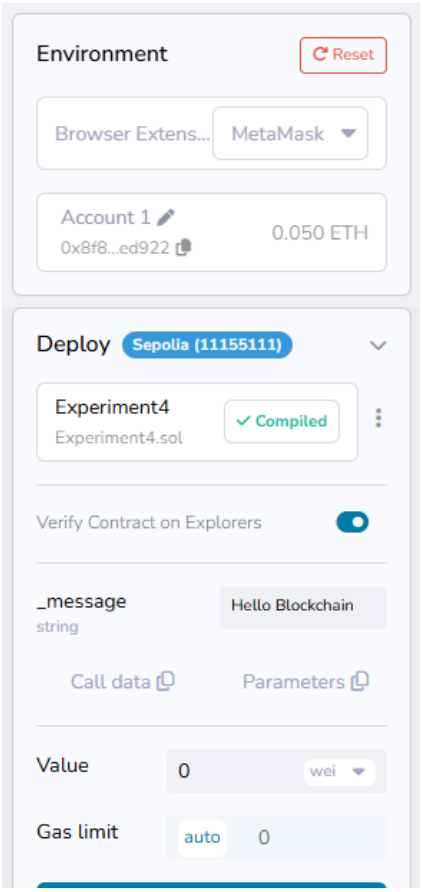
3. Fund Your Wallet from any Testnet:





Step - 2 : Connect the Sepolia Testnet / RSK Testnet to Remix IDE



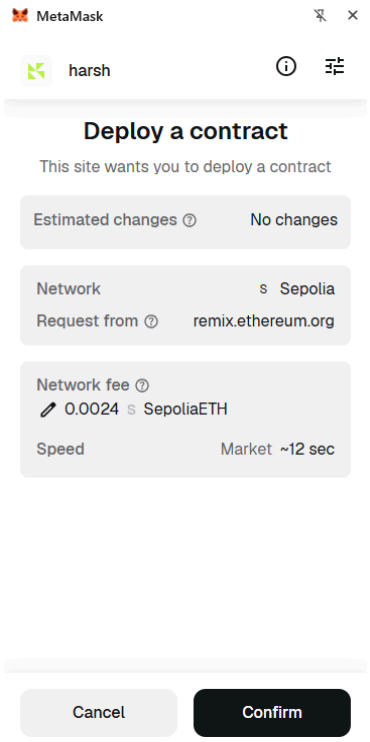


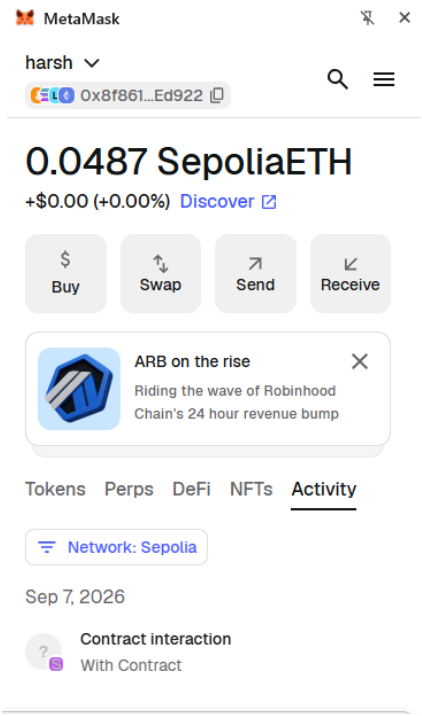
Step - 3 : Create a Simple Solidity Smart Contract



Step - 4 : Compile and Deploy the Smart Contract.

1. Once the Contract is deployed on Remix IDE, the popup appears to confirm the Contract Deployment.





4. Transaction Deployment details are displayed on Sepolia Etherscan w.r.t the Gas Limit, Gas Used, Gas Price and Total Amount.

CONCLUSION:

The smart contract was successfully created, compiled, and deployed using **Solidity and Remix IDE**. MetaMask was successfully integrated with Remix and used to authorize blockchain transactions on the testnet. The experiment provided practical understanding of **smart contracts, MetaMask, testnets, transaction execution, and blockchain interaction.**